# Introduction and Math Review Practice 1

Module 1 - Introduction & Math Review · Daily practice sheet · No date

Name _____ Date _____

**1a** In an experiment to study the chemical decomposition, a drug solution was prepared and a sample was obtained at different time points. The drug concentrations in the samples were as follows.

| Time (hr) | Concentration (mg/L) |
|---|---|
| 2 | 294.3 |
| 6 | 208.1 |
| 12 | 123.8 |
| 24 | 43.8 |
| 36 | 15.5 |
| 48 | 5.5 |

What is the rate constant for the decrease in concentration?

**1b** A drug solution decomposes by a first-order process with a rate constant of 0.0866 $hr^{-1}$. The concentration measured 12 hours after the solution was prepared was 123.8 mg/L. What was the initial concentration of the solution?

**1c** A drug solution decomposes by a first-order process with a rate constant of 0.0866 $hr^{-1}$. How much time is required for exactly half the solution to decompose?

**1d** A drug solution decomposes by a first-order process with a rate constant of 0.0866 $hr^{-1}$. How much time is required for the original solution to decompose by 90 per cent?

**2a** In an experiment to study the chemical decomposition, a drug solution was prepared and a sample was obtained at different time points. The drug concentrations in the samples were as follows.

| Time (hr) | Concentration (mg/mL) |
|---|---|
| 2 | 338 |
| 6 | 314 |
| 12 | 278 |
| 24 | 206 |
| 36 | 134 |
| 48 | 62 |

What is the rate constant for the decrease in concentration?

**2b** A drug solution decomposes by a zero-order process at 6 (mg/mL)/hr, and the sample taken at 2 hours measured 338 mg/mL. What was the initial starting concentration of the solution?

**2c** A drug solution decomposes by a zero-order process at 6 (mg/mL)/hr from an initial concentration of 350 mg/mL. How much time is required for exactly half the solution to decompose?

**2 (final part, unnumbered)**

A drug solution with an initial concentration of 350 mg/mL decomposes by a zero-order process at 6 (mg/mL)/hr. How much time is required for the original solution to decompose by 90%?